SIMATS ENGINEERING

ASSESSMENT TOOL 2: SCENARIO-BASED QUESTIONS

CO2 - AT2 | Question Paper

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Instructions to Students

1. This paper contains 30 scenario-based questions on image enhancement and segmentation.
2. Read each scenario carefully and understand the problem.
3. For every question, answer parts (a) to (e) in your own words.
4. Write clear steps, give reasons for your choice, and draw small diagrams where useful.
5. Write your answers neatly in your assignment notebook or answer sheet.

Question 6: Object Boundary Detection

Scenario: In a medical image, the boundaries between different tissues must be found clearly.

(a) Why is boundary detection important?
(b) What makes it difficult?
(c) Which edge detector can be used?
(d) Why did you choose it?
(e) Write your answer step by step.

Question 11: Low Brightness Image

Scenario: An image taken in low light looks very dark and details are hard to see.

(a) Why is the image so dark?
(b) What problems does it cause?
(c) Which enhancement technique can be used?
(d) Why did you choose it?
(e) Write your answer step by step.

Question 18: Threshold Selection Problem

Scenario: An image needs to be segmented, but it is hard to choose a threshold value.

(a) Why is choosing a threshold difficult?
(b) What issues occur in threshold selection?
(c) Which method can select it automatically?
(d) Why did you choose it?
(e) Write your answer step by step.

Question 6: Object Boundary Detection

Scenario: In a medical image, the boundaries between different tissues must be found clearly.

(a) Why is boundary detection important?
Boundary detection identifies the edges of objects or tissues in an image. It helps separate different regions and identify their shapes.

(b) What makes it difficult?

- Medical images may contain noise.
- Different tissues may have similar intensity values.
- Some boundaries may be weak or unclear.

(c) Which edge detector can be used?
The Canny Edge Detector can be used.

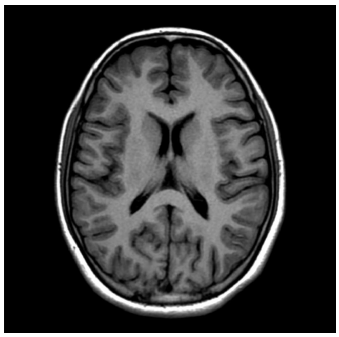
(d) Why did you choose it?
The Canny edge detector reduces noise, detects strong intensity changes, and produces thin, clear edges. Therefore, it is suitable for detecting tissue boundaries.

(e) Step-by-step procedure:

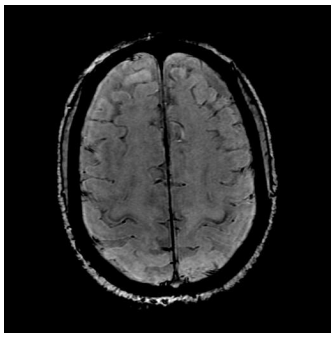
1. Read the medical image.
2. Convert it into grayscale.
3. Apply Gaussian filtering to reduce noise.
4. Calculate the image intensity gradients.
5. Apply non-maximum suppression to obtain thin edges.
6. Use double thresholding and edge tracking to identify connected edges.
7. Display the detected boundaries.

Diagram:

Input Image



Detected Edges



Question 11: Low Brightness Image

Scenario: An image taken in low light looks very dark and details are hard to see.

(a) Why is the image so dark?

The image is dark because insufficient light reaches the camera sensor, resulting in low pixel intensity values.

(b) What problems does it cause?

- Objects and details are difficult to identify.
- Important features may be hidden in dark regions.
- Object detection and segmentation become difficult.

(c) Which enhancement technique can be used?

Histogram Equalization can be used to improve image brightness and contrast.

(d) Why did you choose it?

Histogram equalization redistributes pixel intensity values across a wider range. This can make details in a dark image more visible.

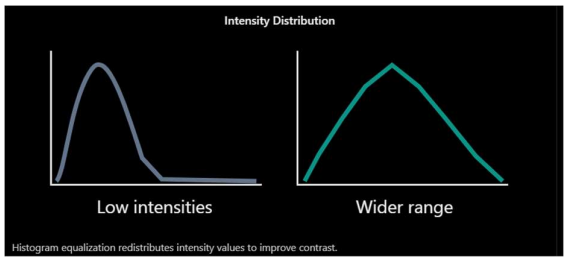
(e) Step-by-step procedure:

1. Read the dark image.
2. Convert the image into grayscale if required.
3. Calculate the histogram of pixel intensities.
4. Apply histogram equalization.
5. Generate the enhanced image.
6. Compare the original and enhanced images to observe the improvement.

Diagram:

Enhanced Image





Question 18: Threshold Selection Problem

Scenario: An image needs to be segmented, but it is hard to choose a threshold value.

(a) Why is choosing a threshold difficult?

Choosing a threshold is difficult when the object and background have overlapping intensity values or when the image has uneven illumination.

(b) What issues occur in threshold selection?

- A low threshold may include unwanted background pixels.
- A high threshold may remove important object pixels.
- Incorrect thresholding produces inaccurate segmentation.

(c) Which method can select it automatically?

Otsu's Thresholding Method can automatically determine a suitable threshold value.

(d) Why did you choose it?

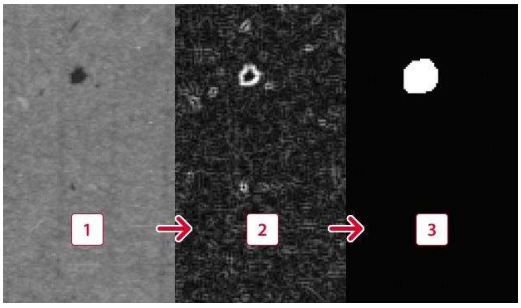
Otsu's method analyses the image histogram and selects a threshold that separates two intensity classes by maximizing the between-class variance. It reduces the need to choose the threshold manually.

(e) Step-by-step procedure:

1. Read the input image.
2. Convert it into grayscale.
3. Calculate the image histogram.
4. Evaluate possible threshold values.
5. Select the threshold using Otsu's method.
6. Classify pixels into foreground and background.
7. Display the segmented image and check the result.

Diagram:

Grayscale Image



Binary Segmentation

